Supporting information for: Two-dimensional powder diffraction in layered van der Waals films: quantitative X-ray area-detector modeling

This supporting information accompanies the manuscript titled *Two-dimensional powder diffraction in layered van der Waals films: quantitative X-ray area-detector modeling* and provides expanded workflow notes and supplemental derivations.

S1 Geometric ray acceptance, detector mapping, and calculation sequence

A weighted incident state supplies a ray origin $\mathbf{r}_0$, unit propagation direction $\hat{\mathbf{s}}_i$, wavelength λ, and source weight w_s. The incident line is

$$\mathbf{r}(u) = \mathbf{r}_0 + u\hat{\mathbf{s}}_i, \qquad u \geq 0. \tag{S1}$$

Let the sample entrance plane pass through $\mathbf{r}_s$ and have outward unit normal $\hat{\mathbf{n}}$. Solving $(\mathbf{r} - \mathbf{r}_s) \cdot \hat{\mathbf{n}} = 0$ gives

$$u_{\text{hit}} = \frac{(\mathbf{r}_s - \mathbf{r}_0) \cdot \hat{\mathbf{n}}}{\hat{\mathbf{s}}_i \cdot \hat{\mathbf{n}}}, \qquad \mathbf{r}_{\text{hit}} = \mathbf{r}_0 + u_{\text{hit}}\hat{\mathbf{s}}_i. \tag{S2}$$

A parallel ray, or a solution with $u_{\text{hit}} \leq 0$, does not reach the sample. For orthonormal in-plane sample directions $\hat{\mathbf{e}}_1$ and $\hat{\mathbf{e}}_2$, define

$$\xi = (\mathbf{r}_{\text{hit}} - \mathbf{r}_s) \cdot \hat{\mathbf{e}}_1, \qquad \upsilon = (\mathbf{r}_{\text{hit}} - \mathbf{r}_s) \cdot \hat{\mathbf{e}}_2. \tag{S3}$$

A finite rectangular support accepts the state only if

$$|\xi| \leq \frac{W}{2}, \qquad |\upsilon| \leq \frac{H}{2}. \tag{S4}$$

The illuminated footprint is the weighted distribution of the accepted (ξ, υ) values. It is fixed by the source distribution and the intersection test rather than applied later as an independent position shift. For the convention in which $\hat{\mathbf{n}}$ points out of the film, the state-specific grazing angle is

$$\theta_i^{(s)} = \sin^{-1}(-\hat{\mathbf{s}}_i \cdot \hat{\mathbf{n}}). \tag{S5}$$

The ordered-film configurations used for the displayed Bi_2Se_3 and Bi_2Te_3 calculations specify an unbounded in-plane sample plane. Equation (S4) documents the available finite-support model, but sample-edge clipping is not invoked as a fitted broadening mechanism in those comparisons.

The detector mapping is most transparent in its own rigid coordinate frame. Let $\mathbf{R}_D$ and $\mathbf{t}_D$ map detector-frame vectors and points into the laboratory frame. For an exit ray that begins at $\mathbf{r}_{\text{hit}}$ and has unit laboratory direction $\hat{\mathbf{s}}_f$, the detector-frame origin and direction are

$$\mathbf{o}_D = \mathbf{R}_D^{\mathsf{T}}(\mathbf{r}_{\text{hit}} - \mathbf{t}_D), \qquad \mathbf{s}_D = \mathbf{R}_D^{\mathsf{T}}\hat{\mathbf{s}}_f. \tag{S6}$$

The detector plane is $z_D = 0$. Its forward intersection is

$$v_D = -\frac{o_{D,z}}{s_{D,z}}, \qquad \mathbf{p}_D = \mathbf{o}_D + v_D\mathbf{s}_D, \tag{S7}$$

provided $s_{D,z}$ is nonzero and $v_D > 0$. The front-face convention additionally requires the exit direction to approach the active face. If (c_0, r_0) is the calibrated reference coordinate and p_c, p_r are the column and row pitches, the continuous detector coordinates are

$$c = c_0 + \frac{p_{D,x}}{p_c}, \qquad r = r_0 + \frac{p_{D,y}}{p_r}. \tag{S8}$$

For a detector with N_c columns and N_r rows, the active-area condition is

$$-\frac{1}{2} \le c \le N_c - \frac{1}{2}, \qquad -\frac{1}{2} \le r \le N_r - \frac{1}{2}. \tag{S9}$$

Detector distance, beam center, and detector tilts are contained in $\mathbf{R}_D$, $\mathbf{t}_D$, and the reference coordinate. Native pixel masses or finite angular-bin quantities are obtained by integrating the continuous detector-coordinate density over the corresponding region. The detector solid-angle factor is already contained in the coordinate Jacobian that defines that density, so it is not multiplied a second time during profile construction.

Reflection identities are constructed before source integration. If $Q_{\max}$ is the largest momentum transfer subtended by the active detector over the wavelength interval and ΔQ is a declared coverage margin, the symbolic candidate set is

$$\mathcal{R} = \left\{(h,k,l) \in \mathbb{Z}^3 : \text{allowed}(h,k,l),\ |\mathbf{G}_{hkl}| \le Q_{\max} + \Delta Q\right\}. \tag{S10}$$

Here allowed imposes the reflection conditions of the structural model. For the rod representation, integer (h, k) identities are enumerated by the same detector-coverage rule and the coordinate along each rod remains continuous.

The production calculation proceeds in the following order:

1. Compile the physical rod identities, systematic absences, structure strengths, sample geometry, detector geometry, and source-state description.

2. Construct weighted source states in two transverse positions, two angular coordinates, and wavelength. The current source sampler uses seeded equal-mass antithetic Latin-hypercube states, so each state carries weight $1/N_s$.

3. Intersect each source ray with the sample plane and apply the selected finite-support test, if any.

4. Transform the accepted incident state into the sample frame and apply the standard planar-interface correction that determines the internal incident propagation state.

5. For each retained rod, treat the mosaic distribution as a continuous density and solve the wavelength-specific Ewald intersections analytically. No separate stochastic crystallite-orientation cloud is deposited on the detector.

6. Map each accepted internal exit state through the planar boundary to its external direction, then evaluate Eqs. (S6)–(S9).

7. Sum the source-weighted continuous detector-coordinate density and integrate it over the declared native pixels or finite angular bins.

Convergence of the finite source representation is checked by increasing the source-state count and, where useful, repeating the seeded construction. The continuous mosaic integral and analytic Ewald roots are part of the deterministic forward measure rather than a second stochastic broadening mechanism. Numerical configurations, source revisions, and profile definitions must remain bound to each retained result.

S2 Crystallographic displacement conventions and current implementation scope

The main text writes the structure factor in the conventional fractional-coordinate form

$$F_{hkl} = \sum_a o_a f_a(Q) T_a(hkl) \exp[2\pi i(hx_a + ky_a + lz_a)] \,. \tag{S11}$$

The displacement factor $T_a(hkl)$ must be interpreted in the crystallographic convention of the reported U or B parameters. For an isotropic mean-square displacement $U_{\mathrm{iso},a}$,

$$T_a(hkl) = \exp\left[-8\pi^2 U_{\mathrm{iso},a} \left(\frac{\sin\theta}{\lambda}\right)^2\right] = \exp\left(-\frac{1}{2} U_{\mathrm{iso},a} Q^2\right), \tag{S12}$$

with $B_{\mathrm{iso}} = 8\pi^2 U_{\mathrm{iso}}$ and $Q = 4\pi \sin\theta/\lambda$.

The executable CIF interface accepts isotropic U_{iso} or B_{iso} values and stores their equivalent isotropic mean-square displacement. General per-site anisotropic CIF metadata are not imported by the current calculation. When directional damping is released in the ordered-film parameterization, one shared uniaxial tensor is defined in the film frame,

$$\mathbf{U}_{\mathrm{film}} = U_R \left(\mathbf{I} - \hat{\mathbf{n}}\hat{\mathbf{n}}^{\mathsf{T}}\right) + U_z \hat{\mathbf{n}}\hat{\mathbf{n}}^{\mathsf{T}}, \tag{S13}$$

which gives

$$T_{\mathrm{dir}}(\mathbf{Q}) = \exp\left[-\frac{1}{2}\left(U_R Q_R^2 + U_z Q_z^2\right)\right]. \tag{S14}$$

This pair is shared by the selected ordered structural model. It is not a site-resolved anisotropic refinement. Any future use of general U_{ij} values must state the CIF tensor convention and the verified transformation into the calculation basis before those values are compared with crystallographic reports.

S3 Wavelength sampling and moving Ewald construction

For wavelength sample λ_j and incident direction $\hat{\mathbf{s}}_{i,j}$, define

$$k_{0,j} = \frac{2\pi}{\lambda_j}, \qquad \mathbf{k}_{i,j} = k_{0,j}\hat{\mathbf{s}}_{i,j}. \tag{S15}$$

In reciprocal-space coordinates centered on the origin, the corresponding Ewald sphere is centered at $-\mathbf{k}_{i,j}$ and has radius $k_{0,j}$. An allowed reciprocal vector contributes only when

$$|\mathbf{k}_{i,j} + \mathbf{G}_{hkl}| = k_{0,j} \tag{S16}$$

within the declared instrumental and numerical resolution.[1] The incident direction is resampled as well when beam divergence is included. Changing wavelength therefore moves the Ewald-sphere center and changes its radius. The calculation reconstructs and tests Eq. (S16) for every wavelength and direction state. A set of concentric shells about a shared center is not the physical wavelength construction.

For one lattice-plane spacing at fixed orientation, Bragg's law gives a useful limiting check,

$$2d\sin\theta_B = \lambda, \qquad d\theta_B = \frac{d\lambda}{\lambda}\tan\theta_B. \tag{S17}$$

This scalar derivative does not replace the full detector-space condition, because sample orientation, incident direction, mosaicity, refraction, and detector acceptance remain coupled. It is used only as a local check on the radial direction in which wavelength changes the detector position.

The wavelength assignment of the Bi_2Se_3 003 streak is tested with an otherwise-identical control. The full detector quantity is

$$I_{\text{full}}(x,y) = \sum_j w_j I(x,y;\lambda_j), \qquad \sum_j w_j = 1, \tag{S18}$$

where every wavelength state has its own incident wavevector and Ewald intersection. The control replaces the distribution by its weighted mean,

$$I_{\text{mono}}(x,y) = I(x,y;\bar{\lambda}), \qquad \bar{\lambda} = \sum_j w_j\lambda_j, \tag{S19}$$

while holding the geometry, structure factors, continuous mosaic density, background treatment, and detector mapping fixed.

The full-spectrum calculation reproduces the long radial extension through 003. The mean-wavelength control collapses that extension, while the transverse ϕ width remains because it is set primarily by the crystallite-orientation distribution. The 006 feature lies at approximately twice the radial 2θ coordinate of 003 and remains more localized in the measured image. The stronger apparent 003 signal is not attributed to $|F_{003}|^2$ alone. Because 003 lies closer to the strong low-Q_z specular response, the wavelength convolution samples a higher-intensity radial envelope than it does around 006. The detector calculation therefore accounts for both the structure strength and the wavelength-resolved Ewald, Jacobian, and intensity weights before the two features are compared.

This result does not imply a universal inverse-order bandwidth law. The radial extension is a property of the full source distribution and detector measure for the stated geometry. The transverse ϕ width is treated separately as the cleaner mosaic observable.

S4 Staged refinement workflow (expanded)

1. **Beam characterization (direct beam).** Acquire direct-beam images at several detector distances. Fit the spot position and shape versus distance to determine the incident-beam widths (w_x, w_z) and Gaussian divergences $(\Delta\theta_x, \Delta\theta_z)$ that set the instrument-limited broadening.

2. **Detector geometry calibration (3D powder standard).** Measure a polycrystalline hBN calibrant and subtract the detector dark frame. Fit multiple Debye–Scherrer rings in detector coordinates and refine detector distance, detector tilts (β, γ), in-plane detector rotation χ_D, and beam center (x_0, y_0).

3. **Sample alignment and goniometer misalignment.** For each dark-subtracted sample image, use indexed Bragg-peak centers rather than full-image intensity. With detector parameters fixed, refine sample alignment by minimizing detector-space residuals between measured and calculated peak positions spanning both specular and off-specular regions. Refine $(\theta_i, \chi, z_B, z_S)$ and the goniometer-axis misalignment rotations (α, ψ). Include film dimensions (W, H) and thickness t to weight footprint and absorption.

4. **Mosaicity/profile refinement.** With geometry fixed, compare selected local detector ROIs or local $I(\phi)$ profiles after center alignment. The shared profile model is a wrapped two-component angular distribution with a Gaussian-like core and a Lorentzian-like broad component. It is not a pseudo-Voigt common-width peak approximation. Off-specular arcs mainly inform the narrow core, while off-condition specular intensity is sensitive to the broad component together with bandwidth, divergence, alignment, and background. The parameterization yields component widths and a broad-component weight, but attribution and necessity require the controlled no-tail and re-optimized Gaussian-only comparisons specified under C18.

5. **Ordered-structure refinement.** With geometry and mosaicity fixed, refine the ordered intensity model against background-subtracted integrated counts in selected detector-space Bragg regions of interest (ROIs) from multiple dark-subtracted images. The corresponding prediction is the calculated detector intensity deposited in those same ROIs. Use one nuisance scale per image and physically constrained ordered-structure parameters.

6. **Disorder refinement.** After the ordered model converges, introduce stacking-disorder parameters only through fixed-in-plane-momentum rod profiles reported versus the film-normal momentum. The measured profile is formed from caked signal and normalization fields by summing signal and normalization over the selected rod mask before division. Geometry, detector calibration, beam terms, mosaicity, lattice constants, and the ordered baseline remain fixed. Only stacking-disorder parameters and declared scale/background terms are refined.

All detector images are dark-subtracted before fitting. Any additional flat-field, polarization, or detector-efficiency correction should be applied consistently to all images or stated explicitly if omitted. The calculated detector-coordinate density already contains the detector-coordinate Jacobian, including the pixel solid-angle factor, so a second solid-angle correction must not be applied to the model output. We recommend stage-specific uncertainty estimation by repeated constrained refits: resample the calibrant points for detector geometry, indexed peak lists for sample alignment, local reflection regions for mosaicity, accepted Bragg ROIs for ordered intensity, and selected rod bins for disorder. Report the standard deviation of the accepted solutions and propagate parameters to the next stage only after the preceding stage is stable.

S4.1 Coordinate remapping, projected profiles, and uncertainty propagation

The detector image remains the primary data object. The 2θ–ϕ representation is an overlap-weighted remapping used to evaluate fixed-m or fixed-Q_R trajectory masks and projected Q_z or L profiles. It is not treated as an independent measurement. The same remapping, masks, and profile extraction are applied to measured and calculated detector images.

The caked image is constructed with a sparse detector-to-cake operator

$$M \in \mathbb{R}^{B\times P}, \tag{S20}$$

where P is the number of detector pixels, B is the number of caked bins, and M_{bk} is the fractional overlap weight from detector pixel k into caked bin b. For detector signal s_k and normalization n_k,

$$S_b = \sum_k M_{bk} s_k, \qquad N_b = \sum_k M_{bk} n_k, \qquad I_b = \frac{S_b}{N_b}, \tag{S21}$$

for bins with $N_b > 0$. Signal and normalization are therefore accumulated before division. This prevents a projected profile from becoming a sum of already normalized pixels and keeps the observable tied to the detector-space calibration.

For each remapped bin, the calibrated detector and sample geometry define a scattering vector $\mathbf{Q}$. The coordinates used to label selected trajectories are

$$\mathbf{Q}_\parallel = \mathbf{Q} - (\mathbf{Q}\cdot\hat{\mathbf{n}})\hat{\mathbf{n}}, \qquad Q_R = |\mathbf{Q}_\parallel|, \qquad Q_z = \mathbf{Q}\cdot\hat{\mathbf{n}}, \qquad L = \frac{Q_z}{|\mathbf{c}^*|} = \frac{Q_z c}{2\pi}, \tag{S22}$$

where $\hat{\mathbf{n}}$ is the film normal. For a hexagonal basal plane, the radial family index is $m = h^2{+}hk{+}k^2$. The index m is a label, not a multiplicity. A selected fixed-Q_R or fixed-m family is retained when

$$|Q_R - Q_{R,0}| \le \Delta Q_R, \qquad Q_{z,\min} \le Q_z \le Q_{z,\max}. \tag{S23}$$

This region is simple in the physical Q_R–Q_z description but appears curved in the displayed 2θ–ϕ map because the coordinate transformation is nonlinear. The curved shape is therefore a consequence of the projection, not an additional physical model.

For a selected profile coordinate ℓ, let $r_{\ell b}$ be the mask or interpolation weight that assigns caked bin b to that profile point. The profile is

$$I_\ell^{\mathrm{ROI}} = \frac{\sum_b r_{\ell b} S_b}{\sum_b r_{\ell b} N_b} = \frac{\sum_k a_{\ell k} s_k}{\sum_k a_{\ell k} n_k}, \qquad a_{\ell k} = \sum_b r_{\ell b} M_{bk}. \tag{S24}$$

Each point in the plotted profile is therefore an average over a finite detector-space band. Its coordinate may be labeled by Q_z or L, but the point represents a finite-bin projection rather than the intensity at one exact reciprocal-space coordinate.

The transformed-coordinate uncertainty has two parts: uncertainty in the coordinate label and uncertainty in the rebinned intensity. The coordinate-label uncertainty follows from the finite detector-pixel footprint and from uncertainty in the detector calibration. For square detector pixels with pitch p, detector distance d, and $t = 2\theta$, the leading flat-detector terms are

$$\sigma_{2\theta,\mathrm{pix}}(t) = \frac{p\cos^2 t}{\sqrt{12}\,d}, \qquad \sigma_{\phi,\mathrm{pix}}(t) = \frac{p}{\sqrt{12}\,d\tan t}. \tag{S25}$$

The radial term is finite and largest near the direct beam. The azimuthal term scales as $1/\tan(2\theta)$, so the problematic regime is confined to the near-specular, low-2θ region where azimuth is poorly conditioned.

For the $d = 75$ mm and $p = 300\,\mu$m geometry used in the uncertainty calculation, the maximum radial coordinate uncertainty is approximately 0.066°. The azimuthal coordinate uncertainty falls below 1° above $2\theta \approx 3.8°$, below 0.5° above $2\theta \approx 7.5°$, and below 0.25° above $2\theta \approx 14.9°$. More generally, the threshold for a target azimuthal width w_ϕ is

$$2\theta_{\mathrm{crit}} = \arctan\left(\frac{p}{\sqrt{12}\,d\,w_\phi}\right), \tag{S26}$$

with w_ϕ expressed in radians.

Additional calibration uncertainty can be propagated with the usual Jacobian form. If the uncertain geometry parameters are collected in $\mathbf{g}$ with covariance matrix Σ_g, then

$$\sigma^2_{2\theta,\mathrm{cal}} = J_{2\theta}\Sigma_g J_{2\theta}^{\mathsf{T}}, \qquad \sigma^2_{\phi,\mathrm{cal}} = J_\phi \Sigma_g J_\phi^{\mathsf{T}}. \tag{S27}$$

If the displayed bin-center label is also treated as uncertain, the finite bin width contributes

$$\sigma^2_{2\theta,\mathrm{bin}} = \frac{\Delta(2\theta)^2}{12}, \qquad \sigma^2_{\phi,\mathrm{bin}} = \frac{\Delta\phi^2}{12}. \tag{S28}$$

These coordinate terms describe the uncertainty of the transformed axis value. They are separate from intensity uncertainty.

The intensity uncertainty is inherited from the detector through the same overlap weights used to construct the caked image and the projected profile. If v_k is the detector-pixel variance and the detector-pixel variances are independent, then

$$\mathrm{Var}\left(I_\ell^{\mathrm{ROI}}\right) = \frac{\sum_k a_{\ell k}^2 v_k}{\left(\sum_k a_{\ell k} n_k\right)^2}. \tag{S29}$$

If overlapping interpolation weights are used, neighboring profile points can be statistically correlated. The corresponding covariance is

$$\mathrm{Cov}\left(I_\ell^{\mathrm{ROI}}, I_q^{\mathrm{ROI}}\right) = \frac{\sum_k a_{\ell k} a_{qk} v_k}{\left(\sum_k a_{\ell k} n_k\right)\left(\sum_k a_{qk} n_k\right)}. \tag{S30}$$

Thus the same weights that define the projected profile also propagate the detector-pixel variance. The covariance between neighboring projected points is neglected in the present line-shape comparison unless formal profile error bars are required.

The displayed coordinate for a projected point can be represented by the same finite-band weighting. For example,

$$\bar{L}_\ell = \frac{\sum_b r_{\ell b} N_b L_b}{\sum_b r_{\ell b} N_b}, \qquad \sigma^2_{L,\ell} = \frac{\sum_b r_{\ell b} N_b (L_b - \bar{L}_\ell)^2}{\sum_b r_{\ell b} N_b}, \tag{S31}$$

before adding calibration, incidence-angle, wavelength, divergence, and mosaic contributions. This width is a finite-bin projection effect. It is not a counting-statistical error bar.

These projection uncertainties are coupled to model-parameter correlations. Peak positions are sensitive to detector distance, beam center, detector tilt, sample height, and incidence angle. Profile widths are sensitive to both beam divergence and the Gaussian mosaic core. Off-condition profiles are sensitive to the broad mosaic component together with bandwidth, divergence, alignment, and background. Near-critical intensity additionally depends on the optical response and is excluded from evidence for the broad component. Relative integrated intensities are sensitive to scale, footprint, absorption, thickness, and structure-factor parameters. The staged workflow separates these effects by fixing or constraining the parameters tied to peak position before fitting profile widths, and by fixing the geometry and mosaic state before interpreting ordered intensities.

The practical consequences are fourfold. First, the 2θ–ϕ maps and Q_R-selected Q_z profiles are controlled projections of the detector image, not independent ideal reciprocal-space scans. Second, transformed-coordinate uncertainty is concentrated near the low-angle/specular region. Third, intensity variance is propagated through the same overlap weights used for caking and profile extraction. Fourth, the strongest remaining systematic coupling occurs near low L, where direct-beam background, bandwidth, divergence, incidence angle, and optical response overlap. That near-critical region is excluded from evidence for the broad mosaic component.

S4.2 Wrapped mosaic distribution

In the main text we describe out-of-plane mosaicity by a misorientation density $\omega(\Delta\theta)$ in the crystallite tilt angle $\Delta\theta$ relative to the substrate normal. Since $\Delta\theta$ is an orientation angle, $\Delta\theta$ and $\Delta\theta + 2\pi$ represent the same physical state. Accordingly, ω is treated as a 2π-periodic probability density on $\Delta\theta \in (-\pi, \pi]$.

The profile used in the present parameterization is a wrapped Gaussian plus a wrapped Lorentzian,

$$\omega_{\mathrm{w}}(\Delta\theta; p_{\mathrm{tail}}, \Gamma, \sigma) = p_{\mathrm{tail}}\, L_{\mathrm{w}}(\Delta\theta; \Gamma) + (1 - p_{\mathrm{tail}})\, G_{\mathrm{w}}(\Delta\theta; \sigma), \qquad 0 \le p_{\mathrm{tail}} \le 1.$$

This is not a pseudo-Voigt constraint: the Lorentzian half-width Γ and Gaussian standard deviation σ are independent. The wrapped Lorentzian and wrapped Gaussian are

$$L_{\mathrm{w}}(\Delta\theta; \Gamma) = \sum_{q=-\infty}^{\infty} \frac{1}{\pi} \frac{\Gamma}{(\Delta\theta + 2\pi q)^2 + \Gamma^2} = \frac{1}{2\pi} \frac{\sinh\Gamma}{\cosh\Gamma - \cos\Delta\theta},$$

$$\begin{aligned} G_{\mathrm{w}}(\Delta\theta; \sigma) &= \sum_{q=-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} \exp\left[-\frac{(\Delta\theta + 2\pi q)^2}{2\sigma^2}\right] \\ &= \frac{1}{2\pi}\left[1 + 2\sum_{n=1}^{\infty} e^{-n^2\sigma^2/2} \cos(n\Delta\theta)\right]. \end{aligned}$$

Both components integrate to unity over $(-\pi, \pi]$, so ω_{w} is normalized. The reported Lorentzian FWHM is 2Γ, and the reported Gaussian FWHM is $2\sqrt{2\ln 2}\,\sigma$. The local small-width limit recovers the usual line-shape widths, but the refinement and normalization are defined by the wrapped functions above. The broad functional form is phenomenological. These equations define the two-component model. They do not establish its necessity relative to a re-optimized Gaussian-only model.

For a reflection with $Q \equiv |\mathbf{G}_{hkl}|$, mosaicity redistributes intensity in polar angle ϑ with line density $I_0\,\omega_{\mathrm{w}}(\vartheta - \vartheta_B)\,d\vartheta$, where ϑ_B is the Bragg position for that reflection. Random in-plane rotations about the film normal revolve this line density about the Q_z axis, yielding an axially symmetric surface density $\rho(\vartheta)$. With area element $dA = 2\pi Q^2 \sin\vartheta\, d\vartheta$, conservation of intensity gives

$$\rho(\vartheta) = \frac{I_0}{2\pi Q^2 \sin\vartheta}\,\omega_{\mathrm{w}}(\vartheta - \vartheta_B), \qquad 0 < \vartheta < \pi,$$

as used in the main text.

S4.3 Wave propagation, refraction, and absorption

Let $z \geq 0$ denote normal depth into the film and use the field convention $\exp[i(\mathbf{k}\cdot\mathbf{r} - \omega t)]$. For wavelength sample λ_j with spectral weight w_j,

$$k_0(\lambda_j) = \frac{2\pi}{\lambda_j}, \qquad \tilde{n}_q(\lambda_j) = 1 - \delta_q(\lambda_j) + i\beta_q(\lambda_j), \tag{S32}$$

where q identifies the layer. The decrement δ_q controls refraction and critical-angle behavior, whereas β_q controls absorption and penetration depth. Both are wavelength-dependent material inputs. Their source, interpolation, and association with the sampled spectrum are recorded with the calculation configuration.

Tangential matching and the complex normal component.

Translational invariance along a flat interface conserves the tangential wavevector. For a field entering layer q from the ambient medium,

$$\mathbf{k}_{\parallel,q}(\lambda_j) = \mathbf{k}_{\parallel,0}(\lambda_j), \qquad k_{z,q}^{(+)}(\lambda_j) = \sqrt{[\tilde{n}_q(\lambda_j)k_0(\lambda_j)]^2 - \left|\mathbf{k}_{\parallel,0}(\lambda_j)\right|^2}. \tag{S33}$$

The root is chosen so that the field decays in its propagation direction. For the inward solution and the depth convention above, $\operatorname{Im} k_{z,q}^{(+)} \geq 0$.[2] The reciprocal exit field is obtained from the same boundary condition with its own conserved tangential component and propagation direction. Real transmitted angles may be reported after this complex-vector calculation, but they are not obtained by applying Snell's law only to $\operatorname{Re}\tilde{n}_q$.

Interface amplitudes and normal flux.

For a scalar field at an interface from medium p to medium q, the field transmission amplitude is

$$t_{pq} = \frac{2k_{z,p}}{k_{z,p} + k_{z,q}}. \tag{S34}$$

Equation (S34) is an amplitude coefficient, not a transmitted-flux fraction. In the nonabsorbing scalar limit, the associated normal-flux ratio includes the factor $\operatorname{Re} k_{z,q}/\operatorname{Re} k_{z,p}$ in addition to $|t_{pq}|^2$. Absorbing or polarization-resolved layers require the corresponding Poynting-flux or Fresnel normalization. The calculation must therefore declare whether each interface weight is a field amplitude, an intensity, or a normal flux. The quantity $|t|^2$ alone is not generally transmitted flux.

Normal-depth attenuation.

Let $\kappa_i(\lambda_j)$ and $\kappa_f(\lambda_j)$ be the nonnegative imaginary normal components for propagation from the surface to a scattering depth and from that depth back to the surface, respectively:

$$\kappa_i(\lambda_j) = \operatorname{Im} k_{z,i}^{(+)}(\lambda_j), \qquad \kappa_f(\lambda_j) = \operatorname{Im} k_{z,f}^{(+)}(\lambda_j), \qquad \kappa_i, \kappa_f \geq 0. \tag{S35}$$

Here $k_{z,f}^{(+)}$ is the inward reciprocal exit-channel root used by reciprocity. Thus κ_f is the positive attenuation constant for the physical path from depth back to the surface. Equivalently, $\kappa_f = -\operatorname{Im} k_{z,f}^{(\mathrm{out})}$ if an outward wavevector is stored explicitly. For a normalized scattering-depth distribution $\rho(z)$ over a film of thickness t, the propagation weight is

$$W_{\mathrm{depth}}(\lambda_j) = \int_0^t \rho(z) \exp[-2\,(\kappa_i(\lambda_j) + \kappa_f(\lambda_j))\,z]\,dz, \qquad \int_0^t \rho(z)\,dz = 1. \tag{S36}$$

For uniform scattering density, this becomes

$$W_{\mathrm{depth}}(\lambda_j) = \frac{1}{t}\int_0^t \exp[-2\,(\kappa_i(\lambda_j) + \kappa_f(\lambda_j))\,z]\,dz \tag{S37}$$

$$= \frac{1 - \exp[-2\,(\kappa_i(\lambda_j) + \kappa_f(\lambda_j))\,t]}{2\,(\kappa_i(\lambda_j) + \kappa_f(\lambda_j))\,t}. \tag{S38}$$

Equations (S36) and (S38) use normal depth. Multiplying $\operatorname{Im} k_z$ by an oblique ray-path length would count the angular projection a second time. The depth weight is combined with consistently normalized entrance and exit interface factors before it multiplies the structure-factor and mosaic weights.

Detector accumulation over the bandwidth.

For each accepted source-state and rod contribution, the wavelength-resolved detector density is formed with the same geometry, structure-factor, mosaic, detector, and optical interfaces. The detector images are then added incoherently,

$$I_{\mathrm{det}}(x, y) = \sum_j w_j\, I_{\mathrm{det}}[x, y; \lambda_j, \{\tilde{n}_q(\lambda_j)\}]\,. \tag{S39}$$

Changing λ_j changes the moving Ewald construction, the layer optical constants, the complex normal components, and the interface and depth weights. The planar-interface correction is retained for every wavelength state. Its role is expected to be strongest when an incident or exit angle approaches the critical-angle regime. No separate optics sensitivity or optics-off comparison is used in the present argument because the optical correction is not a principal contribution of the manuscript.

S4.4 Parratt-to-kinematic crossover for the specular profile

The detector-derived $m = 0$ profile spans two limiting regimes. The low-Q_z response is described by standard Parratt reflectivity, while the higher-Q_z Bragg response is described by a finite-stack kinematic term. The two branches are matched smoothly over a declared overlap interval. We refer to this matching as the Parratt-to-kinematic crossover. It is a practical connection between two limiting calculations, not a new scattering theory. The main-text figure slot is reserved for a direct-array comparison that places the low-Q_z region and the 003 peak on one coordinate transformation, one background treatment, and one intensity scale.

The plotted coordinate is

$$x = \frac{Q_z}{Q_c}, \qquad L = \frac{Q_z c}{2\pi} = x\frac{Q_c c}{2\pi}. \tag{S40}$$

Here Q_c is the film critical wavevector and c is the out-of-plane lattice constant. The diagnostic crossover coordinate uses Q_z/Q_c and the corresponding L value defined above.

The low-Q_z branch is the Parratt reflectivity.[3] For a layer j with complex refractive index $\tilde{n}_j = 1 - \delta_j + i\beta_j$, the conserved in-plane wavevector fixes the layer-normal component

$$k_{z,j} = \sqrt{(\tilde{n}_j k_0)^2 - \left|\mathbf{k}_\parallel\right|^2}, \qquad k_0 = \frac{2\pi}{\lambda}. \tag{S41}$$

For the depth and field convention used above, the square-root branch is chosen to give decay in the relevant propagation direction. The Fresnel amplitude at the interface between layers j and $j+1$ is

$$r_j^{(0)} = \frac{k_{z,j} - k_{z,j+1}}{k_{z,j} + k_{z,j+1}}. \tag{S42}$$

If the lower interface has RMS roughness σ_j, this coefficient is damped by

$$r_j = r_j^{(0)} \exp\left[-2k_{z,j}k_{z,j+1}\sigma_j^2\right]. \tag{S43}$$

For a finite layer of thickness d_{j+1} below the interface, the round-trip phase factor is

$$P_{j+1} = \exp(2ik_{z,j+1}d_{j+1}). \tag{S44}$$

Starting from the semi-infinite substrate and recursing upward, the effective reflection amplitude is

$$R_j^{\mathrm{amp}} = \frac{r_j + R_{j+1}^{\mathrm{amp}} P_{j+1}}{1 + r_j R_{j+1}^{\mathrm{amp}} P_{j+1}}. \tag{S45}$$

The optical branch in this diagnostic is

$$R_{\mathrm{P}}(Q_z) = \left|R_0^{\mathrm{amp}}(Q_z)\right|^2. \tag{S46}$$

This branch retains total-reflection behavior, refraction, evanescence, finite penetration depth, roughness damping, and multiple reflection between the film interfaces.

The high-Q_z branch is built from the normalized finite-stack kinematic term

$$S_{\mathrm{HT},0}(L) = \frac{I_{\mathrm{HT}}(L; p=0)}{I_{\mathrm{HT}}(0; p=0)}. \tag{S47}$$

The phase coordinate for this structure term is not the external Q_z near the critical edge. To match the Parratt film phase, the finite-stack coordinate is computed from the real propagating part of the film wavevector,

$$Q_{z,\mathrm{film}} = \mathrm{Re}(2k_{z,\mathrm{film}}), \qquad L_{\mathrm{film}} = \frac{Q_{z,\mathrm{film}} c}{2\pi}. \tag{S48}$$

Below the critical edge this real part may approach zero in the weak-absorption limit, while a complex internal wavevector still describes evanescence and absorption. The unscaled kinematic branch is then

$$M(Q_z) = \frac{S_{\mathrm{HT},0}(L_{\mathrm{film}})}{Q_z^2}. \tag{S49}$$

The Q_z^{-2} factor gives the high-Q_z structural envelope, while the internal coordinate L_{film} keeps the finite-stack fringe phase consistent with the optical film phase.

A multiplicative scale places this kinematic branch on the Parratt intensity scale. The scale is fit in a high-Q_z overlap window, excluding the selected $00L$ neighborhoods used in the diagnostic. With fit set $\mathcal{F}$, the log-median estimate is

$$A = \exp[\mathrm{median}_{i\in\mathcal{F}} \left(\ln R_{\mathrm{P}}(Q_{z,i}) - \ln M(Q_{z,i})\right)], \tag{S50}$$

with the default overlap window $5 < Q_z/Q_c < 10$. The scaled kinematic branch is

$$R_{\mathrm{H}}(Q_z) = AM(Q_z). \tag{S51}$$

The crossover window is selected by comparing the two positive finite branches in log intensity,

$$r(x) = \log_{10}\left[\frac{R_{\mathrm{H}}(x)}{R_{\mathrm{P}}(x)}\right]. \tag{S52}$$

Candidate points must lie in the search range $3 \le x \le 10$ and satisfy $|r(x)| \le 0.10$. The selected window $[x_1, x_2]$ is the widest contiguous eligible region with width at least 1.0 in Q_z/Q_c. Ties are broken by the smaller median $|r|$. If no acceptable region is found, the fallback window is $x_1 = 3$ and $x_2 = 6$.

Inside the selected window, the interpolation coordinate and weight are

$$t = \frac{x - x_1}{x_2 - x_1}, \qquad w(t) = 6t^5 - 15t^4 + 10t^3, \tag{S53}$$

with t clipped to the interval $[0, 1]$. The final branch is blended in log reflectivity,

$$R_{\mathrm{blend}}(x) = \begin{cases} R_{\mathrm{P}}(x), & x \le x_1, \\ 10^{(1-w)\log_{10} R_{\mathrm{P}}(x) + w\log_{10} R_{\mathrm{H}}(x)}, & x_1 < x < x_2, \\ R_{\mathrm{H}}(x), & x \ge x_2. \end{cases} \tag{S54}$$

A small positive numerical floor is used only to avoid logarithms of nonpositive roundoff values inside the blend window. The resulting curve is exactly Parratt below the selected window, exactly the scaled structure branch above the window, and smooth through the crossover.

For the joint comparison, the low-Q_z interval and the 003 neighborhood are not normalized independently. The measured detector profile, the Parratt branch, the finite-stack branch, and the combined curve must share one declared background treatment and one intensity scale. The 003 peak serves as the higher-Q_z structural reference. Its radial wavelength broadening is evaluated with Eqs. (S18) and (S19).

S4.5 Detector-derived ordered and disorder observables

The detector-space forward model predicts a continuous detector-coordinate density and its finite-region integrals, while the inverse problems use stage-specific observables. In the ordered-structure stage, the measured quantity for ROI i in exposure e is the background-subtracted detector count

$$y_{ei} = \sum_{(u,v)\in\Omega_{ei}} \left[I_e^{\mathrm{meas}}(u,v) - b_{ei}(u,v)\right],$$

where Ω_{ei} is the selected detector-space Bragg ROI and b_{ei} is the local background model. The corresponding prediction is the detector-coordinate density integrated over the same region,

$$\mu_{ei} = \int_{\Omega_{ei}} I_e^{\mathrm{calc}}(c,r)\, dc\, dr,$$

or the equivalent deterministic sum of native pixel masses. Geometry, source terms, and mosaicity are fixed before this stage. One nuisance scale per image accounts for exposure and flux differences. This defines a multi-image ROI-integrated ordered-intensity objective rather than a full-image pixel objective.

The PbI_2 disorder refinement is a final-stage selected-profile fit, not a second global image refinement. The upstream beam, detector, sample alignment, mosaic/profile parameters, lattice constants, and ordered PbI_2 baseline are fixed before this stage starts. The detector image is first caked into a signal field $S(\phi, 2\theta)$ and a normalization field $N(\phi, 2\theta)$ using the calibrated detector geometry. For the selected-rod conversion, let ϑ_Q denote the polar angle of the scattering vector relative to the film normal. For each caked bin,

$$Q = \frac{4\pi}{\lambda}\sin\left(\frac{2\theta}{2}\right), \qquad Q_R = |\sin\vartheta_Q| Q, \qquad Q_z = \cos\vartheta_Q\, Q.$$

A selected rod with in-plane radius $Q_{R,0}$ is retained when

$$|Q_R - Q_{R,0}| \leq \Delta Q_R, \qquad Q_{z,\mathrm{min}} \leq Q_z \leq Q_{z,\mathrm{max}}.$$

A branch selector may be applied after this reciprocal-space mask so that one branch is used for the default fit and the symmetry-related branch is retained for validation.

Let S_{ij} and N_{ij} denote the caked signal and normalization fields, and let w_{ij} denote the selected-rod mask. The measured profile is

$$I_j^{\mathrm{rod}} = \frac{\sum_i w_{ij} S_{ij}}{\sum_i w_{ij} N_{ij}},$$

for bins with positive selected normalization. Signal and normalization are therefore summed before division. This prevents the final trace from becoming a sum of already normalized pixels and keeps the observable tied to the detector-space calibration.

The model profile is evaluated along the same selected branch. In compact notation,

$$M_j(\mathbf{\Theta}_{\mathrm{dis}}) = B(x_j) + s\left[K * \sum_{p\in\{2H,4H,6H\}} w_p I_p(L; \epsilon_p, N)\right]_j,$$

where x_j is the profile coordinate, L is the corresponding stacking coordinate, B is a low-order background, s is a global scale, K is the fixed finite-profile kernel, and I_p is the finite-stack transition-matrix intensity derived in Sec. S5. When an explicit closed-form broadening kernel is used, its widths are fixed by the upstream resolution and mosaic/profile state or declared before the disorder fit. It is not used to release independent peak areas.

The disorder objective is evaluated only over the retained profile bins,

$$\chi^2_{\mathrm{dis}}(\mathbf{\Theta}_{\mathrm{dis}}, s, B) = \sum_{j\in\mathcal{J}} \rho\left(\frac{I_j^{\mathrm{rod}} - M_j(\mathbf{\Theta}_{\mathrm{dis}})}{\sigma_j}\right),$$

where ρ may be a least-squares or robust loss and $\mathcal{J}$ is the retained Q_z interval. The known Bragg locations on the selected branch are inherited from the fixed detector geometry and ordered model. Peak amplitudes are not fit independently before the stacking model is applied. Otherwise, the diffuse between-peak intensity would already have been absorbed by the preprocessing.

S5 Transition-matrix derivation for PbI_2 stacking disorder

This section gives the PbI_2-specific state construction and the finite-stack intensity used in the main text. The general strategy–propagating layer-pair probabilities with a transition matrix–follows the established theory of one-dimensionally disordered crystals.[4–7] The particular six states, allowed interface events, and fault templates below are the model assumptions introduced for the present PbI_2 analysis.

S5.1 Trilayer amplitudes and six-state basis

The basal registries are

$$0 = (0,0), \qquad 1 = \left(\frac{1}{3}, \frac{2}{3}\right), \qquad 2 = \left(\frac{2}{3}, \frac{1}{3}\right). \tag{S55}$$

Taking the Pb plane as the local origin and writing the Pb–I vertical offset as $\eta = \Delta z_{\mathrm{I-Pb}}/c$, the two orientation-dependent trilayer amplitudes are

$$F_+(h,k,L) = f_{\mathrm{Pb}} + f_{\mathrm{I}} \exp\left\{2\pi i\left[\frac{2h+k}{3} + L\eta\right]\right\} + f_{\mathrm{I}} \exp\left\{2\pi i\left[\frac{h+2k}{3} - L\eta\right]\right\}, \tag{S56}$$

$$F_-(h,k,L) = f_{\mathrm{Pb}} + f_{\mathrm{I}} \exp\left\{2\pi i\left[\frac{2h+k}{3} - L\eta\right]\right\} + f_{\mathrm{I}} \exp\left\{2\pi i\left[\frac{h+2k}{3} + L\eta\right]\right\}. \tag{S57}$$

Atomic form factors retain their usual Q dependence. The two amplitudes satisfy $F_-(h,k,L) = F_+(h,k,-L)$.

For the forward registry step $\mathbf{s} = (1/3, 2/3)$, define

$$\omega = \exp[2\pi i(hs_x + ks_y)] = \exp\left[\frac{2\pi i}{3}(h+2k)\right]. \tag{S58}$$

The ordered state basis and its scattering-amplitude vector are

$$\mathcal{S} = [0F_+, 1F_+, 2F_+, 0F_-, 1F_-, 2F_-], \qquad \mathbf{g} = \begin{pmatrix} F_+ & \omega F_+ & \omega^2 F_+ & F_- & \omega F_- & \omega^2 F_- \end{pmatrix}^{\mathsf{T}}. \tag{S59}$$

Because $\omega^3 = 1$, registry 2 is also one backward step from registry 0.

S5.2 Direction-resolved interface law

The five mutually exclusive interface events are

$$\boldsymbol{\theta} = (a, b_+, b_-, d_+, d_-), \qquad a + b_+ + b_- + d_+ + d_- = 1. \tag{S60}$$

The a event preserves registry and orientation. The b_+ and b_- events shift the registry by $+1$ and -1 without changing orientation. The d_+ and d_- events combine an orientation change with the two handed shift rules. A flip without a registry shift is excluded. For a d_+ event, an $F_+ \to F_-$ step advances the registry and the corresponding $F_- \to F_+$ step returns it. The directions are interchanged for d_-. This reversal is what makes a pure d_+ or d_- process a two-layer 4H repeat rather than a registry random walk.

Let

$$S_+ = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}, \qquad S_- = S_+^{-1} = S_+^{\mathsf{T}}. \tag{S61}$$

With rows denoting the current state and columns the next state, define

$$A = aI_3 + b_+S_+ + b_-S_-, \tag{S62}$$

$$B = d_+S_+ + d_-S_-, \tag{S63}$$

$$C = d_+S_- + d_-S_+. \tag{S64}$$

The full six-state transition matrix is

$$T_6 = \begin{pmatrix} A & B \\ C & A \end{pmatrix}. \tag{S65}$$

For example, the row for the state $0F_+$ is $(a, b_+, b_-, 0, d_+, d_-)$. Every row of T_6 sums to one.

The deterministic parent-event vectors, in the order (a, b_+, b_-, d_+, d_-), are

$$\mathbf{e}_{2H} = (1, 0, 0, 0, 0), \qquad \mathbf{e}_{4H} = (0, 0, 0, 1, 0), \qquad \mathbf{e}_{6H} = (0, 1, 0, 0, 0). \tag{S66}$$

The opposite-handed 4H and 6H parents are obtained by selecting d_- and b_- instead.

S5.3 Exact rod-dependent reduction

The three registry-shift matrices are diagonal in the discrete Fourier basis. In the sector selected by the scattering phase ω, the 6×6 transition reduces exactly to

$$M(\omega) = \begin{pmatrix} a + b_+\omega + b_-\omega^{-1} & d_+\omega + d_-\omega^{-1} \\ d_+\omega^{-1} + d_-\omega & a + b_+\omega + b_-\omega^{-1} \end{pmatrix}. \tag{S67}$$

No registry states are discarded. Equation (S67) is one Fourier block of Eq. (S65). Setting the registry phase to unity gives the orientation-population matrix

$$P = M(1) = \begin{pmatrix} a + b_+ + b_- & d_+ + d_- \\ d_+ + d_- & a + b_+ + b_- \end{pmatrix}. \tag{S68}$$

Unlike $M(\omega)$, which carries complex scattering phases, P is a row-stochastic probability matrix.

S5.4 Finite-stack intensity

Let d be the separation between neighboring trilayer centers and $\xi = \exp(iq_z d)$ the corresponding vertical phase. Define

$$\mathbf{f} = \begin{pmatrix} F_+ \\ F_- \end{pmatrix}, \qquad D_{F^*} = \mathrm{diag}(F_+^*, F_-^*). \tag{S69}$$

If $\boldsymbol{\pi}_0^{\mathrm{ori}}$ is the initial orientation row vector, then the orientation population at trilayer n is

$$\boldsymbol{\pi}_n^{\mathrm{ori}} = \boldsymbol{\pi}_0^{\mathrm{ori}} P^n. \tag{S70}$$

The phase-weighted correlation between trilayers n and $n + \Delta$ is

$$K_{n,\Delta} = \boldsymbol{\pi}_n^{\mathrm{ori}} D_{F^*} M(\omega)^{\Delta} \mathbf{f}. \tag{S71}$$

The ensemble-averaged intensity per trilayer for a finite stack of N trilayers is therefore

$$\begin{aligned} \frac{\langle I_N \rangle}{N} = \frac{1}{N} \sum_{n=0}^{N-1} \boldsymbol{\pi}_n^{\mathrm{ori}} \begin{pmatrix} |F_+|^2 \\ |F_-|^2 \end{pmatrix} \\ + \frac{2}{N} \mathrm{Re} \left[\sum_{\Delta=1}^{N-1} \xi^{\Delta} \sum_{n=0}^{N-1-\Delta} K_{n,\Delta} \right]. \end{aligned} \tag{S72}$$

The first term contains the N self contributions and the second contains all distinct trilayer pairs. The calculation evaluates these finite sums directly, retaining end effects and avoiding singular closed forms when an eigenvalue approaches unity. Equivalent matrix-power and recursive formulations are standard for planar-fault diffraction.[8]

S5.5 Polytype-rich fault templates and population mixture

For each parent-rich population,

$$\boldsymbol{\theta}_j(\epsilon_j) = (1 - \epsilon_j)\mathbf{e}_j + \epsilon_j \mathbf{q}_j, \qquad j \in \{2H, 4H, 6H\}. \tag{S73}$$

The equal-alternative templates used in the present implementation are

$$\mathbf{q}_{2H} = \left(0, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right), \tag{S74}$$

$$\mathbf{q}_{4H} = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, 0, \frac{1}{4}\right), \tag{S75}$$

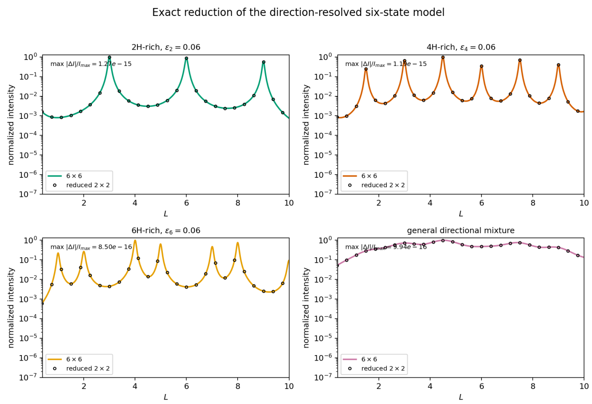
$$\mathbf{q}_{6H} = \left(\frac{1}{4}, 0, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right). \tag{S76}$$

Consequently, ϵ_j is exactly the probability that an interface in population j does not follow its selected parent event. Its numerical value is conditional on the chosen $\mathbf{q}_j$ and must not be interpreted without that template.

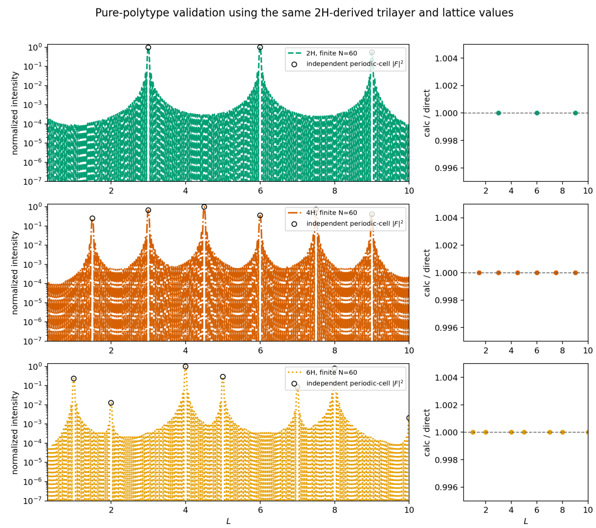
Separate parent-rich populations are not taken to be mutually phase coherent, so their calculated intensities are added:

$$I_{\mathrm{calc}}(L) = S \left\{ \left[w_{2H} I_{2H}(L) + w_{4H} I_{4H}(L) + w_{6H} I_{6H}(L) \right] \otimes K_{\mathrm{prof}}(L) \right\}, \qquad w_{2H} + w_{4H} + w_{6H} = 1. \tag{S77}$$

The core parameterization has three disorder magnitudes and two independent population weights. The stack length N, scale, and any fixed or refined finite-profile and instrumental broadening terms are separate. If the data do not support three independent disorder magnitudes, the constrained starting model is $\epsilon_{2H} = \epsilon_{4H} = \epsilon_{6H}$.



(a) Agreement between the full and reduced calculations.



(b) Recovery of the deterministic parent limits.

Figure S1: Calculation-only validation of the transition construction. The 6×6 and 2×2 expressions agree to numerical precision, and the zero-disorder 2H, 4H, and 6H limits reproduce independently evaluated periodic-parent intensities. These panels validate the implementation. They are not fits to experimental data.

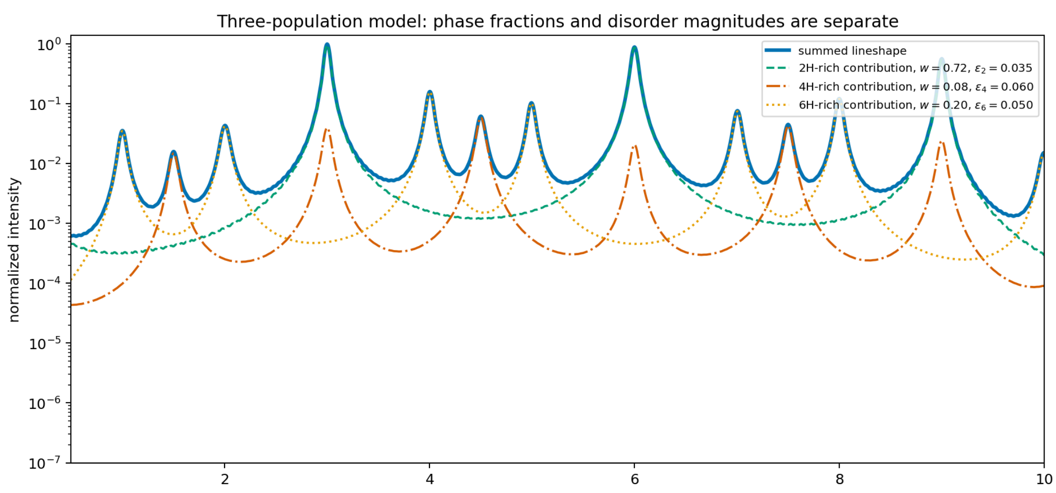


Figure S2: Illustrative incoherent sum of 2H-, 4H-, and 6H-rich intensities. Each population has its own disorder magnitude and one normalized population weight. The numerical values in this calculation-only example are not the experimental fit values.

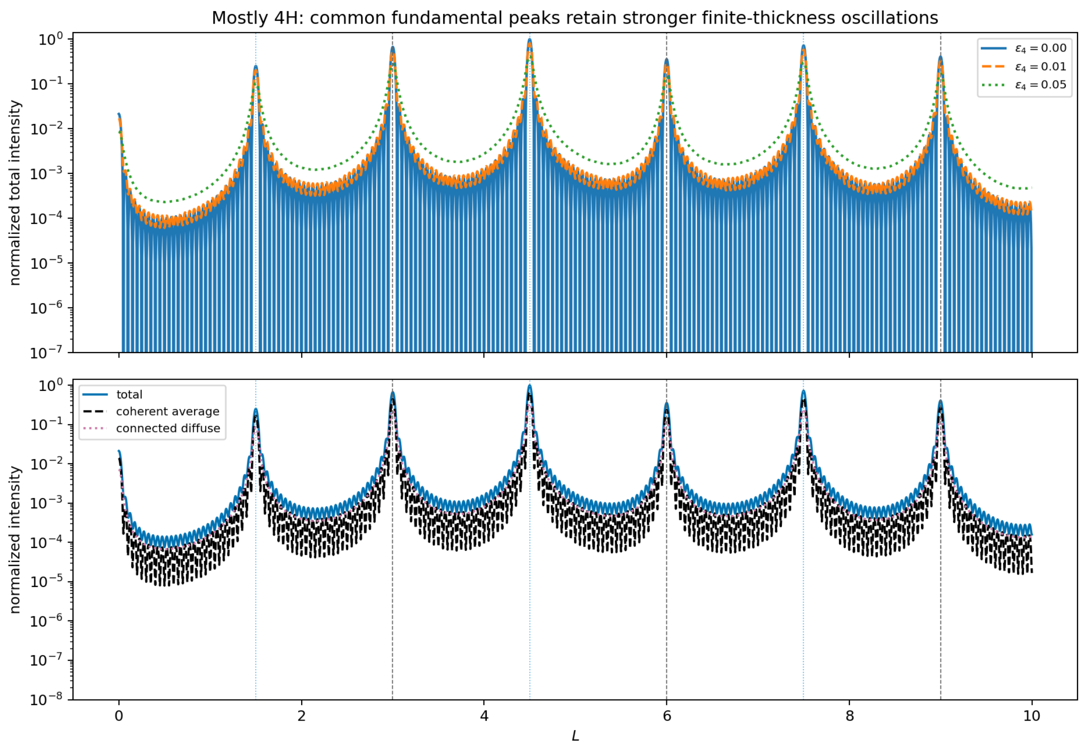


Figure S3: Illustrative sensitivity of common and 4H-specific positions to ϵ_{4H}. Positions shared with the 2H parent retain stronger finite-thickness coherence than positions that require the alternating shift-plus-orientation-change rule.

S5.6 Deterministic and phase-insensitive checks

The zero-disorder cycles are

Parent	Nonzero event	State cycle from $0F_+$	Reduced matrix
2H	$a = 1$	$0F_+ \to 0F_+$	I_2
4H	$d_+ = 1$	$0F_+ \to 1F_- \to 0F_+$	$\begin{pmatrix} 0 & \omega \\ \omega^{-1} & 0 \end{pmatrix}$
6H	$b_+ = 1$	$0F_+ \to 1F_+ \to 2F_+ \to 0F_+$	ωI_2

For 4H, $M(\omega)^2 = I_2$. For 6H, $M(\omega)^3 = I_2$ because $\omega^3 = 1$. The $d_- = 1$ and $b_- = 1$ limits generate the opposite-handed cycles.

For $h = k = 0$, $\omega = 1$ and $F_+ = F_- = F$. Because $P^\Delta \mathbf{1} = \mathbf{1}$, Eq. (S71) becomes $K_{n,\Delta} = |F|^2$ for every allowed transition law. Equation (S72) then reduces to the normalized Laue factor

$$\frac{\langle I_N \rangle}{N} = \frac{|F|^2}{N} \left| \sum_{n=0}^{N-1} \xi^n \right|^2 = \frac{|F|^2}{N} \frac{\sin^2(N\vartheta/2)}{\sin^2(\vartheta/2)}, \qquad \xi = e^{i\vartheta}. \tag{S78}$$

Thus the strong $m = 0$ oscillations are finite-thickness fringes, not diffuse stacking-fault contrast. Faults can still be present. This rod does not distinguish their registry or orientation path.

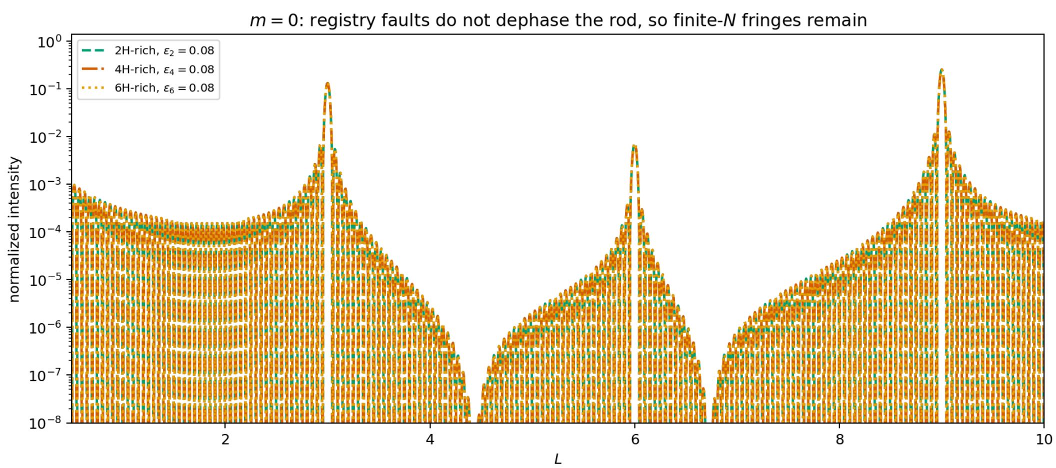


Figure S4: Calculation-only check of the phase-insensitive $m = 0$ condition. Different parent-rich transition laws coincide because registry and orientation changes do not alter the local $00L$ amplitude. The remaining oscillations are controlled by finite stack length.

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